

Bruno Garcia

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EDUCATION

University of Michigan, Ann Arbor, MI

May 2028

Bachelor of Science, Computer Science

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Discrete Math

EXPERIENCE

Cambridge Mobile Telematics

June 2026 – Current

Software Engineer Intern

Cambridge, MA

- Improved performance for the telematics analytics API by caching repeated table initialization every 6 hours, **reducing database calls by 35%** and speeding up dashboard and chart query requests across large customer datasets.
- Built and deployed **MCP tools** with **Terraform** that let an **AWS DevOps agent** query a time-series metrics database for anomalies, reducing root-cause analysis time by **25–50%** for ride failures and device-update issues.
- Built a lightweight **Python polling system** for an analytics chart server that refreshed data every minute, served **1,000+ daily chart requests**, and reduced repeated backend queries by approximately **40%**.
- Built an **anomaly-detection algorithm** in **Python** using **pandas** and **DuckDB** to scan **400,000+ driver records** across rolling 3-day windows and surface ride-failure patterns for further investigation.

Michigan Medicine

May 2025 – Aug 2025

Software Engineer Intern

Ann Arbor, MI

- Standardized **100K+** clinical and medication records from **4 source formats** by building a **Python** ETL pipeline that normalized patient IDs, drug names, dosage units, timestamps, and missing-value conventions.
- Removed **10K+** duplicate records from repeated and overlapping data exports by generating stable record IDs and enforcing idempotent **PostgreSQL** upserts and uniqueness constraints across patient, medication, and visit data.
- Identified **1K+ missing or inconsistent records** by building temporal quality checks in **Python** to flag skipped study visits, incomplete dosing histories, and conflicting timestamps for researcher review.

PROJECTS

[PomodoroLB](#) | JavaScript, Node.js, Next.js, PostgreSQL, Supabase, Vercel

- Built a study app with a live Pomodoro leaderboard adopted by **20+ users**, who logged **100+ study sessions** and **50+ hours**, using Supabase **PostgreSQL**, authentication, cached API queries, and time-windowed SQL rankings.
- Protected leaderboard accuracy by validating authenticated user IDs, blocking invalid and duplicate submissions, and enforcing a **50-minute cooldown** before users could record another Pomodoro session, removing potential cheating.
- Consolidated weekly, monthly, and all-time leaderboard logic into **3 reusable SQL queries**, removing repeated backend logic and simplifying how rankings were calculated as more users submitted sessions.

GlucoFlow | Python, Apache Kafka, PySpark, Redis, PostgreSQL

- Standardized **100K+ CGM readings** across **4+ device formats** by building a **Kafka** and **PySpark** pipeline that normalized timestamps, glucose units, field names, and missing values into a unified PostgreSQL schema for analysis.
- Eliminated **10K+ duplicate readings** from retries and overlapping uploads by generating deterministic record IDs, using **Redis** for atomic checks, and enforcing PostgreSQL uniqueness constraints to keep glucose summaries accurate.
- Identified **1K+ missing intervals** and converted cleaned readings into readable 14-day glucose summaries by calculating time in range, sensor coverage, and data-quality metrics with PySpark and storing the results in PostgreSQL.

SKILLS

Languages: Python, HTML/CSS, SQL, C++, C, R, JavaScript

Tools: Git, SQLite, Docker, BeautifulSoup, REST APIs, JSON, Linux, Tableau, Excel

Libraries: React, Matplotlib, Pandas, scikit-learn

Interests: Rock Climbing, Ping Pong, NBA